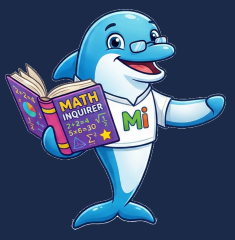
TUTOR GUIDE • LEVEL 1 • TERM 1 • LESSON 10 OF 10

Story Steps: Read It Twice



Big idea: One operation per step. Read a story twice. Then write one number sentence for each step: add, or take away.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

1. At a glance

Strand	Addition and Subtraction (Addition and subtraction within 100 in word problems; Two-step stories: one operation per step; Comparing: more than, fewer than) Strand label for the tutor only. It is not a claim that the program covers the school syllabus, and it must not appear on child or parent pages as 'MOE-aligned'. Lesson 10 is the added tenth lesson of the term (plan, section 2): multi-step stories and reading traps fill a gap found in docs/books/word_problems_g1_g2_analysis.md; the owner or a tutor should check the placement of reversed wording and the who-is-in-the-group trap. Bar strips come in Lesson 12. This lesson also carries the Term 1 end-of-term slot, which is Mi's Challenge Quest (owner decision, 22 Oct 2026: a game, not a test), built as the block challenge_quest by assets/quest_build.py. Term 1 goes beyond the school Primary 1 syllabus on purpose (owner decision, 22 Oct 2026); the stories here use two-digit numbers and two steps with counters and a worked partner step; say so plainly to parents and make no syllabus-match claim. Beyond-syllabus challenge item (owner decision, 22 Oct 2026, every lesson): Think harder T4 is an impossible story (12 stickers, give 7, then give 8): the child must check each step against what is left, a reasoning step beyond the P1 two-step story. Original item.
Objectives	• Read a short story twice: first to say what it is about, then to pick out the numbers and the question. • Write one number sentence for each step of a two-step story (add or take away within 100) and answer in a sentence. • Choose add or take away from the words of the story (gets, gives away, more than, fewer than), not from the order of the numbers. • Play Mi's Challenge Quest, a game with Mi on private paper: 8 short gates that visit the lessons of the term.
Joining cold	**Remember warm-up covers:** 'I think of a number' (add or take away a small number to find it); Story words: which words mean add and which mean take away; Adding and taking away tens and ones: 40 + 20, 37 + 10, 56 - 20, 24 + 5. Only counting to 20, showing a number with fingers or counters, and knowing 10 more or 10 fewer than a tens number. Every number used in Start here stays within 100 and no Start here take away needs renaming. A child who has done Lessons 1 to 9 will find the tens and ones work familiar; a new child gets every idea in the warm-up. Lesson 10 is the last lesson of the term and holds Mi's Challenge Quest, so it is hard to miss: the lesson is recorded, but the quest is not on the recording. A child who missed it plays at the make-up slot, before watching the recording (same private procedure, see the quest tutor notes). A child who joins late is not lost: the warm-up gives what the stories need, the quest is optional for the child, and the line for the grown-up covers only what the child met. Lessons 6 and 7 (jumps of ten, take away) help most for the Start here sums.
Child materials	• About 20 counters: buttons, coins, pasta or the kit squares • Workbook, pencil, rubber, spare paper • Closed Quest Log (4 pages), a ruler with cm, and a quest partner (a grown-up or a brother or sister)

Tutor materials	• A slide deck, and a few counters to act out a story on camera if you wish • The quest tutor notes (L10_Quest_Tutor_Notes.pdf): answers, the private notes sheet for each child and the group advice sheet • Your own watch or clock, out of sight of the camera. No timer or clock is shown during the quest A ruler with centimetres is needed for gate 7 of the quest only; families keep one at home, and there is no printed ruler on the kit page. Nothing needs to be bought: counters can be buttons, coins or pasta. No matches, toothpicks or anything sharp. Print the Quest Log at actual size (100%): the bar at the foot of page 4 must measure 10 cm.
Tiers	Start here (6 questions, pages 1 and 2), Think harder (4, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit page (page 6). Labels are neutral; nobody is told which page they are on.
Contest-style move	Used at Think harder only: two-step story, one operation per step; reversed wording (fewer than, more than); reading trap: who is in the group; three-step story. Never advertised to children or parents.

Before the lesson

- Solve every workbook question and every quest gate yourself first. Read the quest tutor notes (L10_Quest_Tutor_Notes.pdf): the answers, the private notes sheet for each child and the group advice sheet.
- A week before: send the grown-up note (L10_Quest_Parent_Note.pdf) and the Quest Log (L10_Quest_Log.pdf). Families print the Quest Log at 100%, cut out the kit page, find counters, a pencil, a ruler and a quest partner. A family with no printer tells you; arrange another way. The Quest Log stays closed until you say. Build the quest files with: bash term/l1/build.sh 10 then python3 term/l1/assets/quest_build.py 10.
- Have the slides open and your own watch where only you can see it (the deck has no timer slide). Never show a clock, countdown or timer on the shared screen during the quest. Pause the recording for the whole quest and restart it for the look-back.
- Test the recording once with several tiles before the first live class: check that it captures the tutor, the shared screen and any unmuted voice, and not the children's faces. Write the setting you used in your tutor notes.
- Quest photos arrive only on the program's business WhatsApp, +65 9880 5962; open them there, never on a personal phone. Write your private notes from the photos and what you saw on camera, then save the photos to the program's private file store and delete them from the chat. The program keeps them for 90 days after the program ends, then deletes them. Delete each child's notes sheet once the line is sent; keep the group advice sheet for next term's tutor.
- Say plainly if a parent asks: Term 1 goes beyond the school Primary 1 syllabus on purpose. Every Start here item is built to be reached with bars, bundles or a number line and a worked step, and the program does not claim to match any school syllabus. Items that need more than that sit at Think harder.

Safeguarding and recording (this lesson)

- Counters stay on the table: nothing goes in the mouth or ears. No matches, toothpicks or anything sharp. The ruler is for drawing lines only.
- This lesson uses the main room only: no breakout rooms. A parent sits nearby but does not give answers.
- Nobody holds paper or work up to the camera: a gallery view shows everyone. Children tick their own work against the answer slide, answer together on three, or show fingers or a thumb for a choice. Mi's Challenge Quest is the same: papers stay flat on the table, are never held up or read out, and no answer is said aloud.
- Mi's Challenge Quest comes back to the program privately: the parent takes one photo of each page and sends them to the program's business WhatsApp, +65 9880 5962, and to no other number; tutors never use a personal phone for it and never receive it in a chat with the child. The photos are kept for 90 days after the program ends, then deleted (as the privacy notice says). The tutor keeps private notes and sends the parent one line about what we saw. No score is shown to anyone, no ranking, no timer, no comparison between children, no private message to a child.
- Never read out a child's name with a right or wrong answer. The recording captures the tutor, the shared screen and any voice that is not muted. Children may mute and keep their cameras off. Never show a child's page on the recording.
- A child who needs help ticks I am still stuck on their own look-back page. The grown-up may photograph that page and send it to the program's business WhatsApp after class (never to a personal phone); the tutor replies to the grown-up only. The next term starts with one short catch-up question for the whole group. No private messages to children and no private room.
- Movement break: children need a little clear space, soft shoes or socks on a safe floor, no stairs and no running. Lean, do not step backwards. A child may stay seated and move only their arms.
- The look-back is never collected in front of the group or read out. A pass is always allowed, and so is skipping any gate of the quest.

2. Timing plan (90 minutes exactly)

Minutes	Part	Min	Slides	Script fills
0–2	**Opening**	2	1: Story Steps	1:34
2–8	**Remember warm-up**	6	4: Warm-up: I think of a number, Story words, Tens and ones, quickly, Story to number sentence	5:22
8–12	**Sneak-peek puzzles**	4	1: Sneak-peek puzzles	2:56
12–22	**Explore**	10	4: Explore: act it out, Read it twice, Who has more?, What did we notice?	8:45
22–32	**Mini-lesson**	10	8: Read it twice, then step by step, Mei's stickers, Your turn: two steps, Check: two steps, More than, fewer than, Your turn: fewer than, Check: fewer than, Who is in the group?	9:06
32–35	**Brain break**	3	1: Step it out	2:31
35–59	**Workbook**	24	5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page	22:06
59–63	**Talk time**	4	2: Talk and explain, Back to the sneak-peek	3:46
63–83	**Mi's Challenge Quest**	20	2: Mi's Challenge Quest, The Pearl Cave	16:21
83–85	**Look-back**	2	1: Look-back	1:44
85–90	**Flex time and goodbye**	5	1: Well done, inquirers!	1:15
	Total	**90**		**75:27**

The script needs 75:27 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 88:58. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 12 (end of sneak-peek), minute 35 (start of workbook), minute 59 (talk time). If more than 2 minutes behind at minute 35, skip in this order: In a slow class the workbook runs over: take it from the 5-minute flex buffer and shorten choice time (pages 3 to 5 can go home); the optional slide; the second explore story; the quick-fire in the mini-lesson; never shorten Mi's Challenge Quest. Never skip the look-back or the camera and privacy moments.

3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

0:00 – 2:00 Opening

script 1:34 of 2:00

0:00	SLIDE	Show slide **Story Steps** (s-title)
0:25	REC	*If you are watching the recording: hello! Get a pencil, paper, your workbook, counters and a ruler. Pause or rewind any time.*
0:36	SAY	Hello, inquirers! Last lesson of the term. Pencil, paper, counters and your workbook on the table, please. Today we read stories twice, and after our break and our workbook we play Mi's Challenge Quest.
0:53	SAY	This lesson is recorded so a friend who is away can catch up. The recording has me, the shared screen and any voice that is not muted. You may keep your microphone and camera off.
1:10	SAY	Microphone off until I ask you to speak. To answer, say it with me on three, show fingers, or give a thumbs up. Nobody holds paper up to the camera.
1:26	DO	Greet the whole group warmly. Do not use children's first names on the recording. [0:08]
	NOTE	Lesson 10 only: say once that Mi's Challenge Quest is a game on private paper, with a quest partner beside you, and not a race. The routine and the four inquirer rules were taught in Lesson 1; for a new child one sentence is enough: notice it, test it, explain why, mistakes help us. Say the recording statement and the microphone routine in the same words every lesson.

2:00 – 8:00 Remember warm-up

script 5:22 of 6:00

2:00	SLIDE	Show slide **Warm-up: I think of a number** (s-rem1)
2:25	REC	*Recording: join in! Use your fingers or counters, and say each answer before I do. Pause if you need longer.*
2:35	SAY	I think of a number. Use your fingers or counters to find it. I add 3, and I get 8. What is my number?
2:47	ASK	I take away 2, and I get 6. What was my number? [wait about 12 s for answers]
3:05	ASK	I add 10, and I get thirty-five. What was my number? [wait about 12 s for answers]
3:22	SLIDE	Show slide **Story words** (s-rem2)
3:48	SAY	Story words. Gets, finds and buys mean add. Gives away, eats and loses mean take away.
3:56	ASK	Siti eats 3 sweets. Add or take away? [wait about 6 s for answers]
4:06	ASK	Ali finds 5 more shells. Add or take away? [wait about 6 s for answers]
4:16	ASK	Mei gives away 2 stickers. Add or take away? [wait about 6 s for answers]
4:26	ASK	4 more children join the line. Add or take away? [wait about 6 s for answers]
4:38	SLIDE	Show slide **Tens and ones, quickly** (s-rem3)
5:02	SAY	Tens and ones, quickly. Think of bundles of ten.
5:07	ASK	Forty plus twenty? [wait about 8 s for answers]
5:16	ASK	Thirty-seven plus ten? [wait about 8 s for answers]
5:26	ASK	Fifty-six take away twenty? [wait about 8 s for answers]
5:36	ASK	Twenty-four plus five? [wait about 8 s for answers]
5:46	SLIDE	Show slide **Story to number sentence** (s-rem4)
6:10	READ	Story to number sentence. Ben has 6 cards. He gets 4 more. Which number sentence fits: 6 plus 4, or 6 take away 4?

6:24	**ASK**	Which one fits? And what is the answer? [wait about 14 s for answers]
6:42	**ASK**	Siti has 9 beads and loses 3. Which sentence, and the answer? [wait about 14 s for answers]
7:02	**ASK**	Ali has 12 marbles and wins 5. Which sentence, and the answer? [wait about 14 s for answers]
	IF STUCK	If a child hesitates on 47 or 36, say the tens word slowly (forty... seven) and point to the bundles: 37 + 10 is one more bundle.
	NOTE	Answers: 5, 8, 25; take away, add, take away, add; 60, 47, 36, 29; then 6 + 4 = 10, 9 - 3 = 6, 12 + 5 = 17. Children answer with fingers, thumbs or a whispered number. Nobody writes. A child new to tens and ones sees the bundles on the slide first.

8:00 – 12:00 Sneak-peek puzzles

script 2:56 of 4:00

8:00	**SLIDE**	Show slide **Sneak-peek puzzles** (s-sneak)
8:25	**REC**	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*
8:34	**SAY**	Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start. Nobody is judged.
8:44	**READ**	Puzzle one. Ali has 4 marbles. Ben has 3 more marbles than Ali. How many marbles does Ben have? You may draw them.
8:56	**CHILDREN**	Children draw or write the number on their own paper. [0:40]
9:36	**READ**	Puzzle two. How many marbles do Ali and Ben have altogether? Write the number. Then write what you did first.
9:47	**CHILDREN**	Children write or draw their answer. [0:55]
10:42	**SAY**	Thank you. Keep your paper and your own answer: nobody shows it. We will come back to puzzle two later, and you can check your own thinking then.
	NOTE	Puzzle 1 answer: 7. Puzzle 2 answer: 11 (first Ben has 4 + 3 = 7, then 4 + 7). Look for good ideas to praise later (a drawing of two groups, or a child who said Ben first). Do not write scores or keep a list.

12:00 – 22:00 Explore

script 8:45 of 10:00

12:00	**SLIDE**	Show slide **Explore: act it out** (s-explore1)
12:25	**REC**	*Recording: pause now. Act out each story with counters, then say how many. Press play when you have tried.*
12:34	**SAY**	Today your counters are the teachers. I read a story twice. You act it out, one step at a time.
12:44	**READ**	Story A. Mei has 8 buttons. She gets 5 more buttons. Then she gives away 6 buttons. How many buttons does Mei have now?
12:58	**READ**	Mei has 8 buttons. She gets 5 more buttons. Then she gives away 6 buttons.
13:06	**CHILDREN**	Round 1: children act out story A with counters or the kit squares and show a thumbs up when done. [1:10]
14:16	**ASK**	How many buttons does Mei have now? What did you do first? [wait about 20 s for answers]
14:42	**SLIDE**	Show slide **Read it twice** (s-explore2)
15:07	**SAY**	Read it twice. First read: who is in the story and what happens? Second read: which numbers, and what is the question?
15:18	**READ**	Story B. Ravi has 12 stickers. He gives 5 stickers to Siti. Then he finds 3 more stickers. How many stickers does Ravi have now?
15:31	**CHILDREN**	Round 2: children act out story B with counters, one step at a time. [1:00]
16:31	**ASK**	How many stickers does Ravi have now? Which step came first? [wait about 18 s for answers]
16:55	**SLIDE**	Show slide **Who has more?** (s-explore2b)
17:20	**READ**	Story C. Ben has 9 balls. Mei has 4 more balls than Ben. How many balls do they have altogether?

17:31	**READ**	Ben has 9 balls. Mei has 4 more balls than Ben.
17:37	**CHILDREN**	Round 3: children build Ben's pile, then Mei's pile that is 4 more, then put both together. [1:10]
18:47	**ASK**	How many balls does Mei have? How many do they have altogether? [wait about 25 s for answers]
19:18	**SAY**	Mei has more, so Mei has the greater number. Find Mei first. Then put both piles together.
19:26	**SLIDE**	Show slide **What did we notice?** (s-explore3)
19:51	**ASK**	What did you notice? How many steps did each story have? Did each step add or take away? [wait about 35 s for answers]
20:35	**SAY**	Each step is one thing: one add or one take away. Step 2 starts from the answer to Step 1.
	IF STUCK	A child who loses count: put the first number out, then add or remove counters one step at a time and say the new total after each step. A child with too few counters: use the kit squares.
	NOTE	Answers: story A 7 (8 + 5 = 13, then 13 - 6 = 7); story B 10 (12 - 5 = 7, then 7 + 3 = 10); story C: Mei 13, together 22 (9 + 4 = 13, then 9 + 13 = 22). Main room only: no breakout rooms. Building with counters is for every child.

22:00 – 32:00 Mini-lesson

script 9:06 of 10:00

22:00	**SLIDE**	Show slide **Read it twice, then step by step** (s-m1)
22:25	**REC**	*Recording: keep watching and copy the examples. Rewind if you miss a step.*
22:32	**SAY**	Our trick: read it twice, then one number sentence for each step.
22:38	**ASK**	How many number sentences do we write for a story with two steps? [wait about 10 s for answers]
22:54	**SLIDE**	Show slide **Mei's stickers** (s-m2)
23:19	**SAY**	Mei has 26 stickers. Ali gives her 10 more stickers. Then she gives 5 stickers to Ben. Step 1: she gets more, so add. Twenty-six plus ten is thirty-six. Step 2: she gives away, so take away. Thirty-six take away five is thirty-one.
23:40	**ASK**	Where did the thirty-six come from? [wait about 10 s for answers]
23:54	**SLIDE**	Show slide **Your turn: two steps** (s-m3)
24:18	**READ**	Your turn. Siti has 40 beads. She uses 20 beads for a bracelet. Then she buys 7 more beads. How many beads does Siti have now? Write Step 1 and Step 2.
24:36	**CHILDREN**	Children write two number sentences on their own paper. Wait until most have stopped writing. [0:55]
25:31	**SLIDE**	Show slide **Check: two steps** (s-m3a)
25:56	**READ**	Forty take away twenty is twenty. Twenty plus seven is twenty-seven.
26:02	**SLIDE**	Show slide **More than, fewer than** (s-m4)
26:27	**SAY**	More than and fewer than. Ali has 14. Ben has 5 more, so Ben has 19. Ben has 5 fewer, so Ben has 9. Ask: who has more?
26:41	**ASK**	Ben has 5 more than Ali, and Ali has 14. Does Ben have more or fewer? [wait about 10 s for answers]
26:59	**SLIDE**	Show slide **Your turn: fewer than** (s-m4a)
27:24	**READ**	Your turn. Mei has 38 stickers. Siti has 6 fewer stickers than Mei. How many stickers does Siti have? Write the number sentence.
27:36	**CHILDREN**	Children write one number sentence on their own paper. Wait until most have stopped writing. [1:00]
28:36	**SLIDE**	Show slide **Check: fewer than** (s-m4b)
29:02	**READ**	Thirty-eight take away six is thirty-two. Siti has thirty-two stickers.
29:07	**ASK**	Who has more, Mei or Siti? How do you know? [wait about 15 s for answers]
29:27	**SLIDE**	Show slide **Who is in the group?** (s-m5)

29:52	**SAY**	A trap! Ravi is in a line. 5 children are in front of Ravi and 2 are behind him. How many children are in the line? Count Ravi too: 5 plus 2 plus 1 is 8.
30:10	**CHILDREN**	Children line up counters on the table: 5 in front, one for Ravi, 2 behind. They count every counter and show a thumbs up. [0:40]
30:50	**ASK**	Who is in the group? Did we count everyone? [wait about 12 s for answers]
	IF STUCK	If a child reads fewer than as add: ask who has more, and who has the greater number. If a child leaves Ravi out: put counters in a line on the shared slide or whiteboard and touch each one.
	NOTE	Key words to use every time: step, read it twice, more than, fewer than, number sentence. Do not teach bar strips (Lesson 12). Keep every Start here step within 100 and without renaming; the Think harder page has the first renaming.

32:00 – 35:00 Brain break

script 2:31 of 3:00

32:00	**SLIDE**	Show slide **Step it out** (s-break)
32:25	**REC**	*Recording: stand up and join in! A step forward to add, a step back to take away.*
32:34	**SAY**	Stand up and sip some water. Lean forward to add, lean back to take away, in the space you have. Add 3, take away 2, add 4, take away 5. Where do you end up?
32:51	**CHILDREN**	Call the four moves. Children lean forward and back in their chair or standing, in the space they have. Say each count clearly; children count quietly along. Everyone ends back upright: 3, 1, 5, 0. [1:40]

35:00 – 59:00 Workbook

script 22:06 of 24:00

35:00	**SLIDE**	Show slide **Workbook page 1: Start here** (s-ws1)
35:25	**REC**	*Recording: pause now and do workbook page 1. Press play to check it. Then page 2.*
35:33	**READ**	Page 1, Start here. Question 1: Ben has 30 cards. He gives 20 cards away. After that, he gets 6 more cards. Fill in the steps. Question 2: Mei has 42 stickers. She gets 7 more stickers. Then she gives 10 stickers to Ali. Write one number sentence for each step. Then write how many stickers Mei has now. Question 3: Ravi has 25 pencils. He buys 10 more pencils. Then he loses 3 pencils. Circle the two number sentences that fit the story. The four sentences are: 25 plus 10 equals 35; 25 take away 10 equals 15; 35 take away 3 equals 32; 35 plus 3 equals 38.
36:33	**CHILDREN**	Page 1, questions 1 to 3. Tutor looks around the gallery, asks how do you know. Wait 10 seconds before helping. Move to the answer slide only when most children have stopped writing; anyone not finished keeps going and checks later. [7:00]
43:33	**SLIDE**	Show slide **Check page 1** (s-ws1ans)
43:58	**READ**	Answers for page 1. If you are still working, carry on and check later. Question 1: ten, then sixteen. Question 2: forty-nine, then thirty-nine. Question 3: circle 25 plus 10 and 35 take away 3. Ravi has thirty-two. Check your own page in another colour.
44:23	**SLIDE**	Show slide **Workbook page 2: Start here** (s-ws2)
44:48	**READ**	Page 2. Question 4: write plus or minus in each box. Ali has 8 shells and finds 5 more. Siti has 9 sweets and eats 3. Mei has 10 stickers and gives away 4. There are 12 children in a line and 6 more join. Question 5: Ali has 23 marbles. Ben has 10 more than Ali. How many marbles altogether? Question 6: A bus has 30 children. 10 children get off. Then 4 children get on. Siti writes 30 plus 10 equals 40 for Step 1. Is Siti correct? Circle Yes or No. Then find how many children are on the bus now.
45:44	**CHILDREN**	Page 2, questions 4 to 6. Move to the answer slide only when most children have stopped writing. [7:00]
52:44	**SLIDE**	Show slide **Check page 2** (s-ws2ans)
53:09	**READ**	Answers for page 2. If you are still working, carry on and check later. Question 4: plus, minus, minus, plus. Question 5: Ben has thirty-three. They have fifty-six. Question 6: no. Children who get off leave the bus, so take away. Thirty take away ten is twenty. Twenty plus four is twenty-four.
53:38	**SLIDE**	Show slide **Choose your next page** (s-choice)

54:03	SAY	When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can take pages 3 to 5 home.
54:16	CHILDREN	Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. [2:50]
	IF STUCK	Q1: say what happens first; gives away means take away. Q2: use counters, one step at a time. Q3: ask which sentence starts from the answer of the first. Q4: does the word make it greater or smaller? Q5: find Ben first. Q6: ask who leaves the bus.
	IF STUCK	Think harder T1: Ravi has more, so Mei has fewer; draw two groups. T2: draw the line and touch each child, Mei too. T3: find Mei first, then Siti. T4: do step 1, then ask if there is enough for step 2. Go further: write one step at a time.
	IF STUCK	A child who is still stuck ticks I am still stuck on their own look-back page. The grown-up may photograph that page and send it to the business WhatsApp after class. Open the next term with one short catch-up question for the whole group. Never a private message to a child, never a private room.
	NOTE	Think harder is for ideas taught today in a new setting (reversed wording, the group trap). Go further is mastery and extension. Do not announce who is on which page. Pairs are never split unless both children agree. Start here pages 1 and 2 are the target; the rest of the workbook time is for choosing, and pages 3 to 5 may go home.

59:00 – 63:00 Talk time

script 3:46 of 4:00

59:00	SLIDE	Show slide **Talk and explain** (s-talk)
59:25	REC	*Recording: pause. Tell a grown-up your answer to the talk question, and say why.*
59:32	READ	Siti has 20 stickers. Ali has 5 fewer stickers than Siti. A child says Ali has 25 stickers. Do you agree? Use counters to explain.
59:46	CHILDREN	Build 20 and take 5 away with your counters. Is Ravi right? [0:35]
60:21	CHILDREN	Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [0:50]
61:11	ASK	Who would like to share? Volunteers first. A pass is always fine. [wait about 35 s for answers]
61:52	SLIDE	Show slide **Back to the sneak-peek** (s-sneakans)
62:17	SAY	Remember puzzle two? Ben has 3 more than Ali, and Ali has 4. Ben has seven. Together they have eleven.
62:27	ASK	Did anyone change their mind about puzzle two while we worked? You can just nod. [wait about 12 s for answers]
	NOTE	Model answer: No. Fewer means less. Ali has 20 - 5 = 15. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed.

63:00 – 83:00 Mi's Challenge Quest

script 16:21 of 20:00

63:00	SLIDE	Show slide **Mi's Challenge Quest** (s-quest1)
63:25	REC	*Recording: Mi's Challenge Quest is not recorded. Skip ahead. Ask your grown-up how to play it live or at the make-up slot.*
63:36	SAY	Now it is time for Mi's Challenge Quest! Mi the dolphin is swimming to the Pearl Cave, and eight gates are in the way. At each gate there is a puzzle for you. You work on your own paper with your quest partner beside you.
63:58	SAY	It is a game, not a race, and nobody gets a score. You may skip a gate and Mi will swim round it. I will not say any answers, and I will not show anyone's paper.
64:16	DO	Pause the recording. Children take out the closed Quest Log, a pencil and a ruler. Quest partners sit beside the child and may talk about how to start, but do not give answers. Paper flat on the table, never held up. Name on page 1. [0:30]

64:46	**READ**	Page 1. Gate 1. Look at the sticks. How many bundles of ten, and how many single sticks? Write them. Then write how many sticks there are altogether. Gate 2. Fill in the blanks. Part a: look at the number line. Which number is the arrow pointing to? Part b: count on in tens and write the next number: 14, 24, 34.
65:20	**CHILDREN**	Children work quietly on page 1 (gates 1 and 2); partners may whisper about how, not about the answer. Look around the gallery; say nothing about right or wrong. If asked, re-read a gate to the whole group. [1:40]
67:00	**SAY**	Gates one and two are open! Mi swims on. Turn to page 2.
67:07	**READ**	Gate 3. You have two cards, 3 and 8. Use both cards to make the greatest number you can. Write it. Gate 4. Look at the ten frames: 7 and 5. Write 7 plus 5. Then write 37 plus 5. Gate 5. Part a: write 52 plus 30. Part b: write 74 take away 31.
67:37	**CHILDREN**	Children work quietly on page 2 (gates 3 to 5). [2:30]
70:07	**SAY**	Gates three, four and five are open!
70:10	**MOVE**	One-minute stretch: stand, shake out your hands, sip some water. Paper stays on the table.
70:50	**SAY**	Turn to page 3. You need your ruler for gate 7.
70:56	**READ**	Gate 6. Fill in the missing number in each box. Part a: 8 plus 4 equals 6 plus box. Part b: 60 plus 10 equals 50 plus box. Gate 7. Use your ruler. Part a: draw a line 6 centimetres long, starting at the dot. Part b: write how long a line 4 centimetres longer than your line is.
71:28	**CHILDREN**	Children work quietly on page 3 (gates 6 and 7). [2:50]
74:18	**SAY**	Gates six and seven are open! The Pearl Cave is just ahead. Turn to page 4.
74:26	**READ**	Gate 8. Read it twice. Mei has 35 stickers. Ali gives her 10 more stickers. Then she gives 3 stickers to Ben. Write a number sentence for Step 1 and for Step 2. The last number you write is how many stickers Mei has now. The secret cave has one more puzzle, and it is optional. You may skip it. Ali is in a line. There are 5 children in front of him and 3 children behind him. How many children are in the line?
75:12	**CHILDREN**	Children work quietly on page 4: gate 8, then the optional secret cave item if they wish. [2:50]
78:02	**SLIDE**	Show slide **The Pearl Cave** (s-quest2)
78:27	**SAY**	You reached the Pearl Cave! Make sure your name is on page 1 and close your Quest Log. Well done for trying your best. Grown-ups, please take one photo of each page today and send them to the business WhatsApp, as in the note. Mi's tutor will send you one line about what we saw, never a mark.
78:56	**DO**	Say nothing about answers. Remind the group that nothing is added up and no answer is read out. Restart the recording. Children who did not finish may use the flex time at the end. [0:25]
	IF STUCK	A child who says they cannot do a gate: say to the whole group that skipping is fine, Mi swims round it, and they may come back. A child who is upset: the grown-up may stop at any time; do not single the child out. Never give a hint to one child about an answer.
	NOTE	The answers, your private notes sheet for each child, the group advice sheet and the one-line templates for the grown-up are in L10_Quest_Tutor_Notes.pdf. Do not collect or look at any paper during the lesson. Photos arrive only on the program's business WhatsApp and the line goes to the parent only; never a message to a child, and never a personal phone. The exit ticket on workbook page 5 is for home this lesson: the quest takes its place in class.

83:00 – 85:00 Look-back

script 1:44 of 2:00

83:00	**SLIDE**	Show slide **Look-back** (s-look)
83:25	**REC**	*Recording: pause and fill in your look-back box. You can pass.*
83:30	**SAY**	Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you still need help, tick I am still stuck and show your grown-up. You can pass. Nobody reads it out.
83:50	**CHILDREN**	Children fill in the look-back box on page 5. [0:55]

85:00 – 90:00 Flex time and goodbye

script 1:15 of 5:00

85:00	SLIDE	Show slide **Well done, inquirers!** (s-end)
85:25	REC	*Recording: well done for finishing the term! Show a grown-up one thing you found out.*
85:32	SAY	Well done, inquirers, for the whole term. Next term starts with Lesson 11, Who Is Where? Your tutor will tell you the start date.
85:44	SAY	Please have your Lesson 11 workbook with its cards page cut out by a grown-up (the place cards and the number cards), ten small things to put in a row (toys, cups or shoes), a coin or button, a small paper that says FRONT, a pencil and paper. Keep the cards in an envelope.
	SLIDE	Show slide **Go further: what if?** (s-opt) *(optional)*
	SAY	Optional, only if you are on time: Ali has 20 marbles. He wins 6 and loses 4. Does it matter which comes first?
	CHILDREN	Short think time, then take two answers. [1:30]
	SLIDE	Show slide **If we have time: pick one** (s-menu) *(optional)*
	SLIDE	Show slide **Exit ticket** (s-exit) *(optional)*
	READ	Optional exit ticket. Question 1: a bus has 25 children. 10 children get off. Then 3 children get on. How many children are on the bus now? Question 2: Ben has 6 stickers. Mei has 5 more stickers than Ben. How many stickers does Mei have? Question 3: Siti has 20 beads. Ali has 7 fewer beads than Siti. How many beads does Ali have?
	CHILDREN	Children answer E1 to E3 on page 5 on their own. [1:40]
	SLIDE	Show slide **Check the exit ticket** (s-exitans) *(optional)*
	READ	Eighteen. Eleven. Thirteen. Check your own ticket in a different colour.
	SAY	Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each).
	NOTE	This is the standing buffer. A child who needs more time for the quest may use it. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: 22 either way; same answer, order does not matter when both orders can be done. If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on.
86:12	SAY	Goodbye, and thank you to the grown-ups.

3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content. If you are more than 3 minutes ahead at minute 59, use one menu activity then.
- A few children ahead: they may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask 'How do you know?' and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide. In this lesson the buffer may first be needed by children finishing the quest.

Activity	Minutes	You say	Recording cue	Answer
Is it always true?	4 (optional)	Mei says: you can add and take away in any order and get the same answer. Is that always true? Try 20 + 10 - 5, then 20 - 5 + 10, then 4 - 6 + 10.	*Recording: pause. Try 20 + 10 - 5 and 20 - 5 + 10. Then try 4 - 6 first. What do you notice?*	20 + 10 - 5 = 25 and 20 - 5 + 10 = 25: the same. But from 4 you cannot take away 6 first. The answer is the same when both orders can be done.

Make a story puzzle	4 (optional)	Write a two-step story for your grown-up with numbers within 100. Write the question. Keep your own answer secret, then check theirs.	*Recording: pause. Write a two-step story and a question for your grown-up to solve.*	No single answer. Check that each step is one add or one take away and the question matches.
Teach it back	3 (optional)	Pretend a toy has never heard of reading a story twice. Teach it using: Ali has 20 marbles, wins 6 and loses 4.	*Recording: pause. Teach a toy or a grown-up to read a story twice. Use 20, win 6, lose 4.*	First read: who and what. Second read: the numbers and the question. Step 1: 20 + 6 = 26. Step 2: 26 - 4 = 22.

4. Questions, answers, hints and common mistakes

Sneak-peek puzzles (minutes 8 to 12)

Puzzle 1 (access): Ali has 4 marbles. Ben has 3 more marbles than Ali. How many marbles does Ben have? You may draw them.

> **Answer: 7**

Puzzle 2 (think): How many marbles do Ali and Ben have altogether? Write the number. Then write what you did first.

> **Answer: 11. First find Ben's marbles: 4 + 3 = 7. Then add Ali's 4: 4 + 7 = 11.**

Unmarked. Never put answers on boards in front of the group and never read a name with an answer. Reveal the answers where your script says (usually Talk time).

Start here (workbook pages 1 and 2)

S1 **Ben has 30 cards. He gives 20 cards away. After that, he gets 6 more cards.**

(a) Step 1: 30 - 20 = ☐
(b) Step 2: ☐ + 6 = ☐
(c) Ben has ☐ cards now.

> **Answer: Step 1: 30 - 20 = 10. Step 2: 10 + 6 = 16. Ben has 16 cards.**

Hint (say this first): Say what happens first. Gives away means take away. Gets more means add.
Worked solution: First, Ben gives away 20, so we take away: 30 - 20 = 10. Then he gets 6 more, so we add: 10 + 6 = 16. Ben has 16 cards now.
Common mistakes: Adding the 20 (30 + 20) because 20 is in the story.; Using 30 again in Step 2 instead of the 10 left after Step 1.
Time for a middle child: about 2 min.

S2 **Mei has 42 stickers. She gets 7 more stickers. Then she gives 10 stickers to Ali. Write one number sentence for each step.**

(a) Step 1: ___
(b) Step 2: ___
(c) Mei has ___ stickers now.

> **Answer: Step 1: 42 + 7 = 49. Step 2: 49 - 10 = 39. Mei has 39 stickers.**

Hint (say this first): Use counters or draw it. What happens first? Then what happens next?
Worked solution: Step 1: she gets 7 more, so add: 42 + 7 = 49. Step 2: she gives 10 away, so take away: 49 - 10 = 39. Mei has 39 stickers now.
Common mistakes: Using 42 again in Step 2 (42 - 10 = 32).; Adding both numbers (42 + 7 + 10).
Time for a middle child: about 3 min.

S3 **Ravi has 25 pencils. He buys 10 more pencils. Then he loses 3 pencils. Circle the 2 number sentences that fit the story.**

(a) 25 + 10 = 35
(b) 25 - 10 = 15
(c) 35 - 3 = 32
(d) 35 + 3 = 38
Ravi has ___ pencils now.

> **Answer: Circle 25 + 10 = 35 and 35 - 3 = 32. Ravi has 32 pencils.**

Hint (say this first): Buys more means add. Loses means take away. The second step starts from the answer to the first.
Worked solution: Step 1: buys 10 more, so 25 + 10 = 35. Step 2: loses 3, so 35 - 3 = 32. The sentences 25 - 10 = 15 and 35 + 3 = 38 use the wrong sign.
Common mistakes: Circling 25 - 3 = 22 because both numbers are in the story.; Circling a sentence with the wrong sign.
Time for a middle child: about 2.5 min.

S4 **Write + or - in each box.**

(a) Ali has 8 shells. He finds 5 more shells. 8 ☐ 5 = 13
(b) Siti has 9 sweets. She eats 3 sweets. 9 ☐ 3 = 6
(c) Mei has 10 stickers. She gives away 4 stickers. 10 ☐ 4 = 6
(d) There are 12 children in a line. 6 more children join the line. 12 ☐ 6 = 18

Answer: +, -, -, +

Hint (say this first): Does the story word make the number greater or smaller? Gets more makes it greater.
Worked solution: Finds more means add (+). Eats means take away (-). Gives away means take away (-). Join means more, so add (+).
Common mistakes: Choosing + for gives away because there is a number in the sentence.; Choosing - for join.
Time for a middle child: about 2 min.

S5 **Ali has 23 marbles. Ben has 10 more than Ali. How many marbles do they have altogether? Use counters if you like.**

They have ___ marbles altogether.

Answer: Step 1: Ben has 23 + 10 = 33. Step 2: 23 + 33 = 56. They have 56 marbles.

Hint (say this first): Read it twice. First find how many Ben has. Then put Ali and Ben together.
Worked solution: Step 1: Ben has 10 more than Ali, so 23 + 10 = 33. Step 2: altogether means add Ali and Ben: 23 + 33 = 56 marbles.
Common mistakes: Answering 33 (stopping after Step 1).; Adding 23 + 10 and calling that altogether.
Time for a middle child: about 3 min.

S6 **A bus has 30 children. 10 children get off. Then 4 children get on. Siti writes 30 + 10 = 40 for Step 1.**

(a) Is Siti right? Circle Yes or No.
(b) The bus now has ___ children.

Answer: No. Get off means take away: 30 - 10 = 20. Then 20 + 4 = 24. There are 24 children.

Hint (say this first): Who leaves the bus? Do they make the bus fuller or emptier?
Worked solution: Children who get off leave the bus, so Step 1 is a take away: 30 - 10 = 20. Siti added the 10. Then 4 get on, so add: 20 + 4 = 24.
Common mistakes: Circling Yes.; Adding both numbers to 30 (30 + 10 + 4 = 44).
Time for a middle child: about 3 min.

Think harder (page 3): the taught idea in a new setting

T1 **Ravi has 15 marbles. He has 3 more marbles than Mei.**

(a) Mei has ___ marbles.
(b) They have ___ marbles altogether.

Answer: Mei has 12 marbles. Together they have 27.

Hint (say this first): Ravi has more, so Mei has fewer. Draw two groups of counters and compare them.
Worked solution: Ravi has 3 more than Mei, so Mei has 3 fewer than Ravi: 15 - 3 = 12. Together: 15 + 12 = 27 marbles.
Common mistakes: Adding 15 + 3 = 18 because the word more is there.; Answering 12 for the second part.
Time for a middle child: about 4 min.

T2 **Mei is in a line to buy lunch. There are 6 children in front of her and 4 children behind her. How many children are in the line?**

There are ___ children in the line.

Answer: 11 children (6 + 4 + 1, counting Mei).

Hint (say this first): Draw the line of children. Did you draw Mei too?
Worked solution: The children in front are 6 and behind are 4: 6 + 4 = 10. Mei is in the line as well, so 10 + 1 = 11 children.
Common mistakes: Answering 10 and leaving Mei out.
Time for a middle child: about 4 min.

T3 **Ben has 12 marbles. Mei has 5 more marbles than Ben. Siti has 3 fewer marbles than Mei. How many marbles does Siti have?**

Siti has ___ marbles.

> **Answer: Mei has 17. Siti has 14 marbles.**

Hint (say this first): Find Mei first. Then Siti has fewer than Mei.
Worked solution: Step 1: Mei has 5 more than Ben: 12 + 5 = 17. Step 2: Siti has 3 fewer than Mei: 17 - 3 = 14 marbles.
Common mistakes: Taking 3 from 12 (Siti is compared with Mei, not Ben).; Answering 17 (stopping after Step 1).
Time for a middle child: about 4 min.

T4 **Ali has 12 stickers. He gives 7 stickers to Mei. Then he gives 8 stickers to Ben.**

(a) After giving 7 to Mei, Ali has ___ stickers left.
(b) Can Ali give 8 stickers to Ben? Circle Yes or No.

> **Answer: 5 stickers left; No, the story cannot happen.**

Hint (say this first): Do the first step. How many stickers are left? Is that enough for the second step?
Worked solution: Step 1: 12 - 7 = 5, so Ali has 5 stickers left. Step 2 needs 8 stickers, but 5 is less than 8. Ali cannot give Ben 8, so this story can not be true. A story step can be impossible, so check each step.
Common mistakes: Writing 12 - 7 - 8 and getting an answer without noticing 5 is less than 8.; Circling Yes without checking step 2.
Time for a middle child: about 4 min.

Go further (page 4): mastery and extension

G1 **A shop has 60 pencils. It sells 24 pencils on Monday and 15 pencils on Tuesday. On Wednesday, a box of 30 pencils comes in. Write one number sentence for each step.**

(a) Step 1: ___
(b) Step 2: ___
(c) Step 3: ___
(d) The shop has ___ pencils now.

> **Answer: Step 1: 60 - 24 = 36. Step 2: 36 - 15 = 21. Step 3: 21 + 30 = 51. The shop has 51 pencils.**

Hint (say this first): There are three things that happen, so there are three steps. Write one number sentence for each.
Worked solution: Monday: 60 - 24 = 36. Tuesday: 36 - 15 = 21. Wednesday a box comes in, so add: 21 + 30 = 51 pencils.
Common mistakes: Adding 24 and 15 and taking that from 60 but forgetting the box.; Starting each step from 60 again.
Time for a middle child: about 6 min.

G2 **Step 1: 25 + 10 = 35. Step 2: 35 - 8 = 27. Write a story that fits both steps. End your story with a question. Many stories are right.**

> **Answer: Any story where 25 grows by 10 and then 8 are taken away, and the question asks how many now. The answer is 27.**

Hint (say this first): Pick who and what. Step 1 adds 10. Step 2 takes away 8. What could 25 be?
Worked solution: For example: Ali has 25 stickers. He gets 10 more. Then he gives 8 to his sister. How many stickers does Ali have now? 25 + 10 = 35, 35 - 8 = 27.
Common mistakes: A story where the second step starts from 25 instead of 35.; No question at the end.
Time for a middle child: about 6 min.

G3 **Mei, Ali and Ben have 70 shells altogether. Mei has 25 shells. Ali has 10 more shells than Mei. Write one number sentence for each step. How many shells does Ben have?**

(a) Step 1: ___
(b) Step 2: ___
(c) Step 3: ___
(d) Ben has ___ shells.

> **Answer: Ali has 35. Mei and Ali have 60. Ben has 10 shells.**

Hint (say this first): Find Ali first. Then find how many Mei and Ali have together.
Worked solution: Step 1: Ali has 25 + 10 = 35. Step 2: Mei and Ali have 25 + 35 = 60. Step 3: Ben has the rest: 70 - 60 = 10 shells.
Common mistakes: Answering 35 or 60 (stopping early).; Taking only Mei's shells from 70.
Time for a middle child: about 6 min.

Investigation (page 5): Does the order matter?

Easy first step: Start at 20. Add 10, then take away 5. Start again at 20. Take away 5 first, then add 10. Is the answer the same?
Then: Pick your own numbers. Try both orders. Is the answer always the same? When can one order not work?

> **Model answer: Both orders give the same answer (20 + 10 - 5 = 25 and 20 - 5 + 10 = 25) whenever both orders can be done. But the order can matter for what you are allowed to do: from 4 you cannot take away 6 first, though 4 + 10 - 6 = 8 works. So: the answer is the same, but you cannot take away more than you have.**

Look for: A child who tests several numbers, says what they notice, and finds a case where taking away first is not possible (start 4, take away 6).
Common mistakes: Testing only one example and saying always.; Thinking the answer changes because the numbers change places.

Talk and explain (minutes 59 to 63)

Siti has 20 stickers. Ali has 5 fewer stickers than Siti. Ravi says Ali has 25 stickers. Do you agree? Use counters to explain. Sentence starter: I think ... because ...

> **Model answer: No. Fewer means less. Ali has 5 fewer than 20, so Ali has 20 - 5 = 15 stickers. Siti has more than Ali, so 25 cannot be right.**

Look for: A reason that uses the word fewer or less and shows 20 with 5 taken away (not just because 25 is too big). Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

Exit ticket (minutes 63 to 83): 3 questions

E1 **A bus has 25 children. 10 children get off. Then 3 children get on. How many children are on the bus now?**

The bus now has ___ children.

> **Answer: 18 (25 - 10 = 15, then 15 + 3 = 18)**

Worked solution: Get off means take away: 25 - 10 = 15. Get on means add: 15 + 3 = 18.
Common mistakes: Writing 38 (adding the 10).; Writing 15 (stopping after Step 1).

E2 **Ben has 6 stickers. Mei has 5 more stickers than Ben. How many stickers does Mei have?**

Mei has ___ stickers.

> **Answer: 11 (6 + 5)**

Worked solution: Mei has more than Ben, so add: 6 + 5 = 11.
Common mistakes: Writing 1 (taking away).; Writing 5.

E3 **Siti has 20 beads. Ali has 7 fewer beads than Siti. How many beads does Ali have?**

Ali has ___ beads.

Answer: 13 (20 - 7)

Worked solution: Ali has fewer, so take away: 20 - 7 = 13.
Common mistakes: Writing 27 (adding because the number 7 is there).; Writing 7.

Look-back (minutes 83 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (reading the story twice; add or take away; more than, fewer than; starting Step 2 from Step 1; something else); *What helped me* (counters; drawing it; reading twice; I asked for help; I was not stuck); *Next time I will try* (drawing it; counters; checking; asking); *I am still stuck (my grown-up will photograph this page for my tutor)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

5. Home practice and online practice

#	Tier	Question	Answer
H1	Start here	Ali has 28 marbles. He wins 10 more marbles. Then he gives 5 marbles away. Write one number sentence for each step. Step 1: _____ Step 2: _____ Ali has ___ marbles now.	33
H2	Start here	Mei has 14 stickers. Siti has 6 more. How many stickers does Siti have?	20
H3	Start here	A box has 40 pencils. 10 pencils are red and 20 pencils are blue. The rest are green. Write one number sentence for each step. Step 1: _____ Step 2: _____ There are ___ green pencils.	10
H4	Think harder	Ben has 9 sweets. He has 4 fewer than Kumar. How many sweets altogether?	22
H5	Go further	Make up a two-step story with numbers within 100. Write one number sentence for each step. Ask a grown-up to solve it.	Any two-step story with the right sentences. Check each step: the second sentence starts from the answer to the first.

Online practice (`out/l10/practice`, key `t1l10`): 14 items; 12 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

6. Parent line for the term report

Your child is learning to read a story twice and write one step at a time. This week: two-step stories within 100, 'more than' and 'fewer than', and Mi's Challenge Quest.

Term line (plan, section 5): Your child is learning to see numbers as tens and ones and to solve problems one step at a time. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

7. Slide list

#	Time	Slide	On screen	Speaker note
1	0–2	**Story Steps** s-title	TODAY WE WILL: / Read stories twice and write one step at a time. / Then we show what we learned this term.	Lesson 10 of Term 1, the last lesson of the term. 90 minutes. Objective: read a short story twice and write one number sentence for each step. Time plan: opening 0 to 2, warm-up 2 to 8, sneak-peek 8 to 12, explore 12 to 22, mini-lesson 22 to 32, brain break 32 to 35, workbook 35 to 59, talk 59 to 63, Mi's Challenge Quest 63 to 83, look-back 83 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Nobody is told which page they should be on. Tell the group the lesson is recorded (the tutor, the screen and any voice that is not muted) and that a child may keep the microphone and camera off. Say that Mi's Challenge Quest, after the break, is a game on private paper with a quest partner and is not a race.
2	2–8	**Warm-up: I think of a number** s-rem1	Use your fingers or counters / [picture: ten_frame] / I think of a number. I add 3. I get 8. / What is my number? / + 3 = 8 - 2 = 6 + 10 = 35	Say: I think of a number, I add 3, I get 8. Children use fingers or counters to find the number: 5 (8 take away 3). Then: I think of a number, I take away 2, I get 6 (answer 8). Then: I think of a number, I add 10, I get 35 (answer 25). This is the holiday-course game I think of a number, used only as a warm-up; no writing.
3	2–8	**Story words** s-rem2	If the story says | We usually / gets, finds, buys, joins, altogether | add / gives away, eats, loses, leaves, how many left | take away / Add or take away? Say it.	Read the two rows aloud, then ask: Siti eats 3 sweets (take away). Ali finds 5 more shells (add). Mei gives away 2 stickers (take away). 4 more children join (add). Children answer aloud or with a thumb up for add and down for take away. Say that today we also meet the words more than and fewer than.
4	2–8	**Tens and ones, quickly** s-rem3	[picture: bundles 3 tens 7 ones] / 40 + 20 37 + 10 56 - 20 24 + 5 / Say the answers. Think of tens and ones.	Answers: 60, 47, 36, 29. Show 37 as 3 bundles and 7 singles: 37 + 10 is one more bundle, 47. 56 - 20 takes 2 bundles, 36. 24 + 5 adds 5 singles, 29. This is spiral review of adding and taking away tens and ones, so a child who joins cold can do the lesson. Say the tens word slowly: forty-seven.
5	2–8	**Story to number sentence** s-rem4	Ben has 6 cards. He gets 4 more. / 6 + 4 6 - 4 / Which one fits the story?	Read each short story twice and ask which number sentence fits, then the answer. Ben has 6 cards and gets 4 more: 6 + 4 = 10. Siti has 9 beads and loses 3: 9 - 3 = 6. Ali has 12 marbles and wins 5: 12 + 5 = 17. Children answer aloud or show a thumb up for the first sentence and down for the second. No writing yet.
6	8–12	**Sneak-peek puzzles** s-sneak	Ideas we have not learnt yet. A guess is a great start. Nobody is judged. / PUZZLE 1: anyone can try: / Ali has 4 marbles. / Ben has 3 more marbles than Ali. / How many marbles does Ben have? / You may draw them. / PUZZLE 2: think: / How many marbles do Ali and Ben have altogether? / Write the number. Then write what you did first.	Puzzle 1 is an access question: any child can draw 4 marbles and 3 more, and count. Puzzle 2 is the thinking question on the key idea (two steps). Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper and everyone holds the paper to the camera once, at the same moment, after puzzle 2. Do not read answers aloud. Puzzle 1 answer 7; puzzle 2 answer 11, first Ben has 4 + 3 = 7.

7	12–22	**Explore: act it out** s-explore1	Listen once. Picture the story. / Listen again. Put out counters for the first number. / Do each step with your counters. / Say: how many now? / STORY A: / Mei has 8 buttons. She gets 5 more buttons. Then she gives away 6 buttons.	Round 1, about 3 minutes, objects first and no writing. Read story A twice: Mei has 8 buttons. She gets 5 more buttons. Then she gives away 6 buttons. How many buttons does Mei have now? Children act it out with counters or the kit squares: 8 and 5 more is 13, then 6 away leaves 7. Ask for a thumbs up when each child has finished. Answer for the tutor: 7.
8	12–22	**Read it twice** s-explore2	FIRST READ: / Who is in the story? / What happens? / No numbers yet. / SECOND READ: / Which numbers? / What is the question? / One step at a time. / STORY B: / Ravi has 12 stickers. He gives 5 stickers to Siti. Then he finds 3 more stickers.	Round 2, about 3 minutes. Read story B twice, with a first read for who and what, then a second read for the numbers and the question: Ravi has 12 stickers. He gives 5 stickers to Siti. Then he finds 3 more stickers. How many stickers does Ravi have now? Children act it out: 12 - 5 = 7, then 7 + 3 = 10. Answer for the tutor: 10. Praise the two reads, not the speed.
9	12–22	**Who has more?** s-explore2b	STORY C: / Ben has 9 balls. Mei has 4 more balls than Ben. / How many balls do they have altogether? / Find Mei first. Then put them together.	Round 3, about 3 minutes. Read story C twice: Ben has 9 balls. Mei has 4 more balls than Ben. How many balls do they have altogether? Children act it out with counters: a pile of 9 for Ben, then a pile for Mei that is 4 more (13), then put both piles together (22). Answer for the tutor: Mei 13, together 22. Say that Mei has more, so her pile is the greater one. This sets up question 5 in the workbook.
10	12–22	**What did we notice?** s-explore3	How many steps did each story have? / Did each step add, or take away? / Where did Step 2 start?	Take one or two answers from volunteers. Key ideas: each story had two steps; each step was one add or one take away; Step 2 starts from the answer to Step 1. Repeat a child's answer in your own words so the recording picks it up. This sets up the mini-lesson.
11	22–32	**Read it twice, then step by step** s-m1	FIRST READ: / Who? What happens? / SECOND READ: / Which numbers? What is asked? / One step, one number sentence	Name the routine: read it twice, then write one number sentence for each step. First read: who and what, no numbers. Second read: the numbers and the question. Then each step is one add or one take away. Ask: how many number sentences for a two-step story? Answer: two.
12	22–32	**Mei's stickers** s-m2	STORY: / Mei has 26 stickers. Ali gives her 10 more stickers. Then she gives 5 stickers to Ben. How many stickers does Mei have now? / Step 1 | 26 + 10 = 36 / Step 2 | 36 - 5 = 31 / Answer | Mei has 31 stickers.	Think aloud. First read: Mei, stickers, Ali gives, she gives to Ben. Second read: 26, 10 more, 5 to Ben. Step 1: gets more, so add, 26 + 10 = 36 (one more bundle of ten). Step 2: gives away, so take away, 36 - 5 = 31 (no renaming: 6 - 5 = 1). Write the answer in a sentence. Ask: where did 36 come from? Answer: from Step 1.
13	22–32	**Your turn: two steps** s-m3	Write Step 1 and Step 2 on your paper. About 1 minute. / STORY: / Siti has 40 beads. She uses 20 beads for a bracelet. Then she buys 7 more beads. How many beads does Siti have now? / Step 1 | ___ / Step 2 | ___	Read the story twice. Each child writes on their own paper; wait until most children have stopped writing. Do not ask children to hold boards up in front of the group. Answer for the tutor: 40 - 20 = 20, then 20 + 7 = 27.
14	22–32	**Check: two steps** s-m3a	Step 1 | 40 - 20 = 20 / Step 2 | 20 + 7 = 27 / Siti has 27 beads.	Answers: 40 - 20 = 20 and 20 + 7 = 27. Read each answer aloud so a child watching the recording can mark their own work. Ask: which words told us to take away? (uses) And to add? (buys more).
15	22–32	**More than, fewer than** s-m4	The words | The step / Ben has 5 more than Ali. Ali has 14. | 14 + 5 = 19 / Ben has 5 fewer than Ali. Ali has 14. | 14 - 5 = 9 / Ask: who has more?	More than and fewer than compare two people. Ali has 14. Ben has 5 more, so Ben has 14 + 5 = 19. Ben has 5 fewer, so Ben has 14 - 5 = 9. Always ask who has more. Then the person with more has the greater number. Keep it to one minute; the reversed wording (Ravi has more, find Mei) is for Think harder.

16	22–32	**Your turn: fewer than** s-m4a	Who has more? Write the number sentence. About 1 minute. / STORY: / Mei has 38 stickers. Siti has 6 fewer stickers than Mei. How many stickers does Siti have? / Number sentence | ___	Read the story twice. Each child writes on their own paper; wait until most children have stopped writing. Do not ask children to hold boards up in front of the group. Answer for the tutor: 38 - 6 = 32.
17	22–32	**Check: fewer than** s-m4b	Number sentence | 38 - 6 = 32 / Siti has 32 stickers.	Answer: 38 - 6 = 32. Read it aloud so a child watching the recording can mark their own work. Ask: who has more? (Mei) So who has the smaller number? (Siti). Fewer means take away.
18	22–32	**Who is in the group?** s-m5	STORY: / Ravi is in a line. There are 5 children in front of Ravi and 2 behind him. How many children are in the line? / Count Ravi too! / 5 + 2 + 1 = 8	A reading trap. Show the line on a shared whiteboard, a blank slide or with counters: 5 children, then Ravi, then 2 children. 5 + 2 = 7 leaves Ravi out, so add 1 more: 8 children. Ask: who is in the group? Always check you counted everyone, including the person in the story. This is the taught idea for the Think harder page.
19	32–35	**Step it out** s-break	Lean forward for add. Lean back for take away. / + 3 - 2 + 4 - 5 / Where do you end up? Sip some water.	Three minutes. Stand up on camera. Call each move; children lean forward and back in the space they have (no stepping backwards): add 3, take away 2, add 4, take away 5. Say each count clearly; children count quietly along. After the four moves, everyone is back at the chair: 3, 1, 5, 0. No equipment; keep it gentle for small spaces.
20	35–59	**Workbook page 1: Start here** s-ws1	Questions 1 to 3. About 7 minutes / questions S1, S2, S3	Read questions 1 to 3 aloud, twice. Children work on their own; the tutor looks around the gallery and asks how do you know. Wait 10 seconds before helping. After question 3 any child may carry on or try Think harder. Question 1 shows the steps; question 2 asks for the two sentences; question 3 asks which sentences to circle.
21	35–59	**Check page 1** s-ws1ans	answers S1, S2, S3	Read each answer aloud so a child watching the recording can mark their own work. Question 1: 10, 10 and 16, 16. Question 2: 49 then 39. Question 3: circle 25 + 10 = 35 and 35 - 3 = 32; Ravi has 32. Never ask a child to show a wrong answer to the group.
22	35–59	**Workbook page 2: Start here** s-ws2	Questions 4 to 6. About 7 minutes / questions S4, S5, S6	Read question 4 aloud: write + or - for each story. Read question 5 aloud, twice: Ben has more than Ali, so find Ben first. Read question 6 aloud: it is a spot-the-mistake. Children may draw counters to decide. Children who finish page 2 may choose Think harder, Go further or the Investigation, or finish at home.
23	35–59	**Check page 2** s-ws2ans	answers S4, S5, S6	Question 4: plus, minus, minus, plus. Question 5: Ben has 33, together 56. Question 6: Siti added the 10, but children who get off leave the bus; the right steps are 30 - 10 = 20 and 20 + 4 = 24. Read each answer aloud. Never ask a child to show a wrong answer to the group.
24	35–59	**Choose your next page** s-choice	Page | What it is / Think harder (page 3) | Our idea in a new setting / Go further (page 4) | Longer stories / Investigation (page 5) | Does the order matter? / Check | Finish and check your own work / Everyone may choose. Pages 3 to 5 can go home.	Invite the whole group at once: when you have finished Start here you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. The workbook time after Start here is for choosing; these pages can also be finished at home, and model answers are in the tutor guide.
25	59–63	**Talk and explain** s-talk	A CHILD SAYS: / Siti has 20 stickers. / Ali has 5 fewer stickers than Siti. / A child says Ali has 25 stickers. / Do you agree? Use counters to explain. / I think … because …	One minute of talk to the grown-up next to you or to your screen. Then volunteers first; a pass is always fine; never call a child by name to answer. Model answer: no. Fewer means less, so Ali has 20 - 5 = 15. Repeat each child's answer in your own words for the recording.

26	59–63	**Back to the sneak-peek** s-sneakans	Ben has 3 more than Ali, and Ali has 4. / Step 1 | Ben has 4 + 3 = 7 / Step 2 | Together 4 + 7 = 11 / 11 marbles altogether.	Reveal puzzle 2 now. Say: some of you may have found a good first step. Do not read out anyone's earlier answer. The first step is finding Ben's marbles, 4 + 3 = 7; the second is putting them together, 4 + 7 = 11.
27	optional	**Exit ticket** s-exit	Optional. On your own. No looking back. About 3 minutes / questions E1, E2, E3	OPTIONAL, only if the whole class has finished the quest and you are on time. A short exit ticket: read each question aloud twice, children write on page 5, then check their own ticket on the next slide. Nobody holds the ticket up to the camera. Skip it if you are behind: the quest ends with a two-step story, so nothing is lost.
28	optional	**Check the exit ticket** s-exitans	answers E1, E2, E3	Read each answer aloud. Children check their own ticket in a different colour and may show fingers for how many they got right. The exit ticket is never part of Mi's Challenge Quest and never shown to the group.
29	63–83	**Mi's Chal-lenge Quest** s-quest1	Your own paper. Mi swims with you. / Open your Quest Log. Name on page 1. / Pencil and ruler ready. / Your quest partner sits beside you. / I read each gate. Then you write. / You may skip a gate. It is not a race. / AFTER: / A grown-up takes a photo of each page and sends it to the program's business WhatsApp. / Nothing is shown to the group.	Introduce the quest for about a minute: Mi the dolphin swims to the Pearl Cave through 8 gates and an optional secret cave (about 20 minutes), on your own private paper with a quest partner beside you; a game, not a race; you may skip any gate; no answers are said aloud; nothing is shown to the group; no clock or timer is shown. Pause the recording first. Children take out the closed Quest Log only now. The paper lies flat on the table. Quest partners sit beside the child and may talk about how, but do not give answers. Follow the quest tutor notes (L10_Quest_Tutor_Notes.pdf).
30	63–83	**The Pearl Cave** s-quest2	Make sure your name is on page 1. / Close the Quest Log. Put your pencil down. / A grown-up takes one photo of each page, flat and from above. / A grown-up sends the photos to the program's business WhatsApp number in the grown-up note, and nowhere else. / WELL DONE: / You reached the Pearl Cave! Mi will tell your grown-up one thing we saw.	Say thank you. The grown-up takes one clear photo of each of the 4 pages, with the whole page in view and the name on each page, and sends them today to the program's business WhatsApp, as in the grown-up note. The tutor never asks a child for the sheet and never comments on a child's work in front of the group. Remind the group that no answers will be read out and nothing is added up. Children who did not finish may use the flex time at the end. Restart the recording for the look-back. The number is in the grown-up note, never on a slide.
31	83–85	**Look-back** s-look	[look-back box]	Two minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never graded, never collected in front of the group and never read out. If a child ticks I am still stuck, the grown-up may photograph the page and send it to the business WhatsApp after class; open the next term with one catch-up question for the whole group; never a direct message.
32	85–90	**Well done, inquirers!** s-end	Show a grown-up one thing you found out this term. / NEXT TERM: / Term 2, Lesson 11: Who Is Where? / Cut out the cards on the Lesson 11 workbook kit page and keep them in an envelope. / Bring 10 small things, a coin, a paper that says FRONT, a pencil and paper.	Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. A child who needs longer to finish the quest may use it; otherwise finish any open item. Do not start anything new. If the whole class is ahead, show the optional menu slide and use the Spare-time menu from the tutor guide. Remind families that the tutor sends one line about what we saw in Mi's Challenge Quest, never a mark.

33	optional	**Go further: what if?** s-opt	Ali has 20 marbles. He wins 6 and loses 4. / Does it matter which comes first? / Win first | 20 + 6 - 4 / Lose first | 20 - 4 + 6	Optional, only if you are on time. Win first: 20 + 6 = 26, 26 - 4 = 22. Lose first: 20 - 4 = 16, 16 + 6 = 22. Same answer. Take two answers and move on.
34	optional	**If we have time: pick one** s-menu	Activity | What we do / Is it always true? | Does the order of the steps matter? / Make a story puzzle | Write a two-step story for your grown-up / Teach it back | Teach a toy to read a story twice / Optional. Everyone may join.	Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum.

8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

Chapter	Part	You say (recording voice)	A child watching later
0:00	Opening	*If you are watching the recording: hello! Get a pencil, paper, your workbook, counters and a ruler. Pause or rewind any time.*	Gather your workbook, pencil, paper, counters and a ruler. Press play when ready.
2:00	Remember warm-up	*Recording: join in! Use your fingers or counters, and say each answer before I do. Pause if you need longer.*	Play along. Say each answer out loud before the video does. Pause if you need longer.
8:00	Sneak-peek puzzles	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*	Pause. Try both puzzles on your own paper. A guess is fine.
12:00	Explore	*Recording: pause now. Act out each story with counters, then say how many. Press play when you have tried.*	Pause. Act out each story with counters, one step at a time, and say your answer.
22:00	Mini-lesson	*Recording: keep watching and copy the examples. Rewind if you miss a step.*	Watch and copy the worked examples. Rewind any step you missed.
32:00	Brain break	*Recording: stand up and join in! A step forward to add, a step back to take away.*	Stand up and step it out: forward to add, back to take away.
35:00	Workbook	*Recording: pause now and do workbook page 1. Press play to check it. Then page 2.*	Pause. Do page 1, then play to check. Then page 2. Pages 3 to 5 are for any time.
59:00	Talk time	*Recording: pause. Tell a grown-up your answer to the talk question, and say why.*	Pause. Explain your answer to someone at home, with a reason.
63:00	Mi's Challenge Quest	*Recording: Mi's Challenge Quest is not recorded. Skip ahead. Ask your grown-up how to play it live or at the make-up slot.*	Skip ahead. Play Mi's Challenge Quest live with your group, or at the make-up slot your tutor arranges.
83:00	Look-back	*Recording: pause and fill in your look-back box. You can pass.*	Pause. Fill in the look-back box, or pass.
85:00	Flex time and goodbye	*Recording: well done for finishing the term! Show a grown-up one thing you found out.*	Show a grown-up your workbook and tell them one thing you found out this term.

Do not record these slides. Pause the recording (or show a blank slide) while the quest is on screen, so the shared recording does not give away the questions or answers: s-quest1 (Mi's Challenge Quest); s-quest2 (The Pearl Cave). Say once to the camera: the recording pauses now for Mi's Challenge Quest, and starts again for the look-back.

9. Timing check (model, not a stopwatch)

Part	Allocated	Script	Slow class	Flag
opening	2:00	1:34	1:46	ok
remember	6:00	5:22	6:21	OVER (slow class)
sneak	4:00	2:56	3:26	ok
explore	10:00	8:45	10:20	OVER (slow class)
mini	10:00	9:06	10:45	OVER (slow class)
break	3:00	2:31	2:59	ok
work	24:00	22:06	26:12	OVER (slow class)
talk	4:00	3:46	4:28	OVER (slow class)
exit	20:00	16:21	19:13	ok
look	2:00	1:44	2:03	OVER (slow class)
flex	5:00	1:15	1:25	ok
Total	90:00	75:27	88:58	

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 6 questions; a middle child needs about 7.5 min for page 1 and 8.0 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.